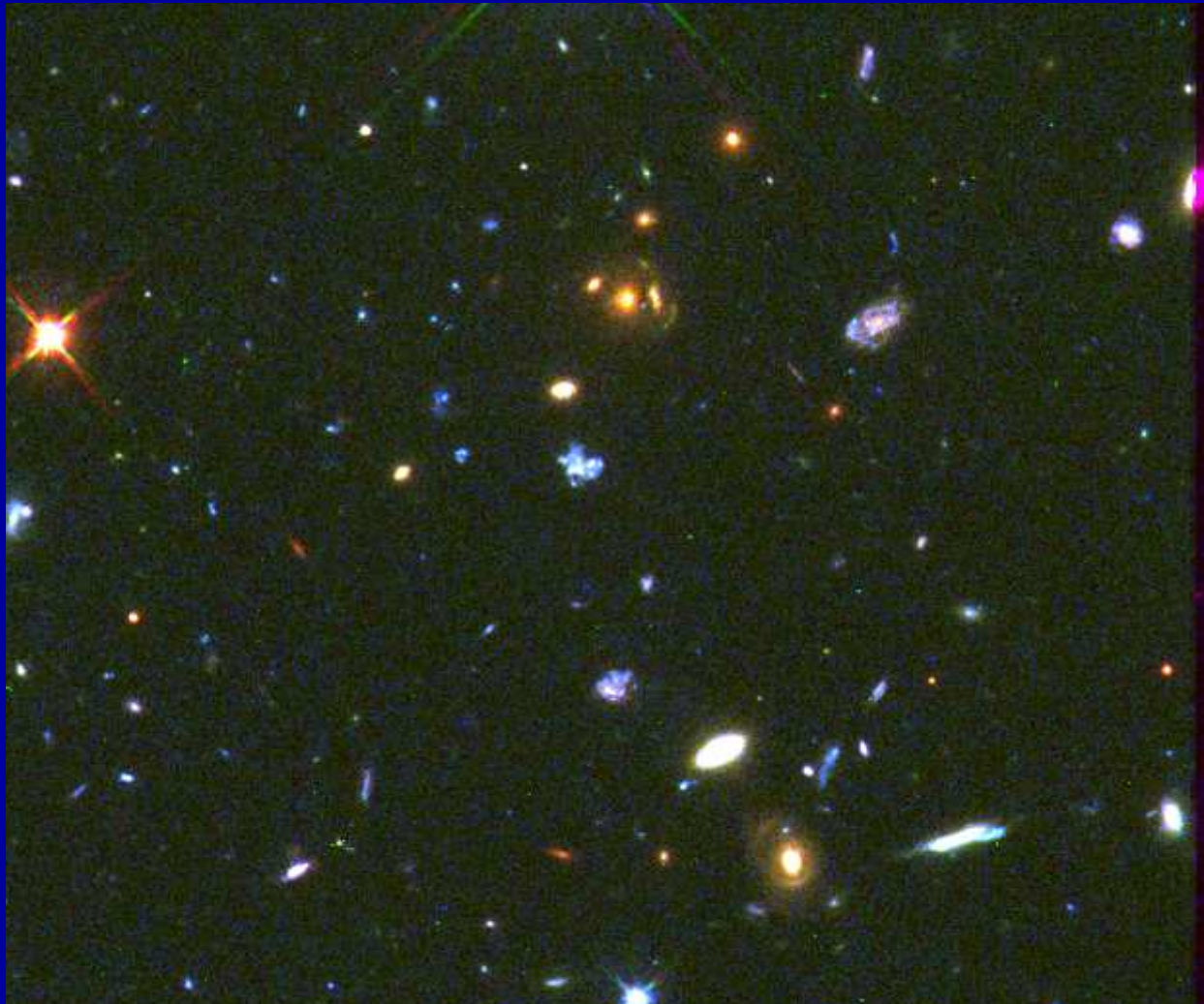


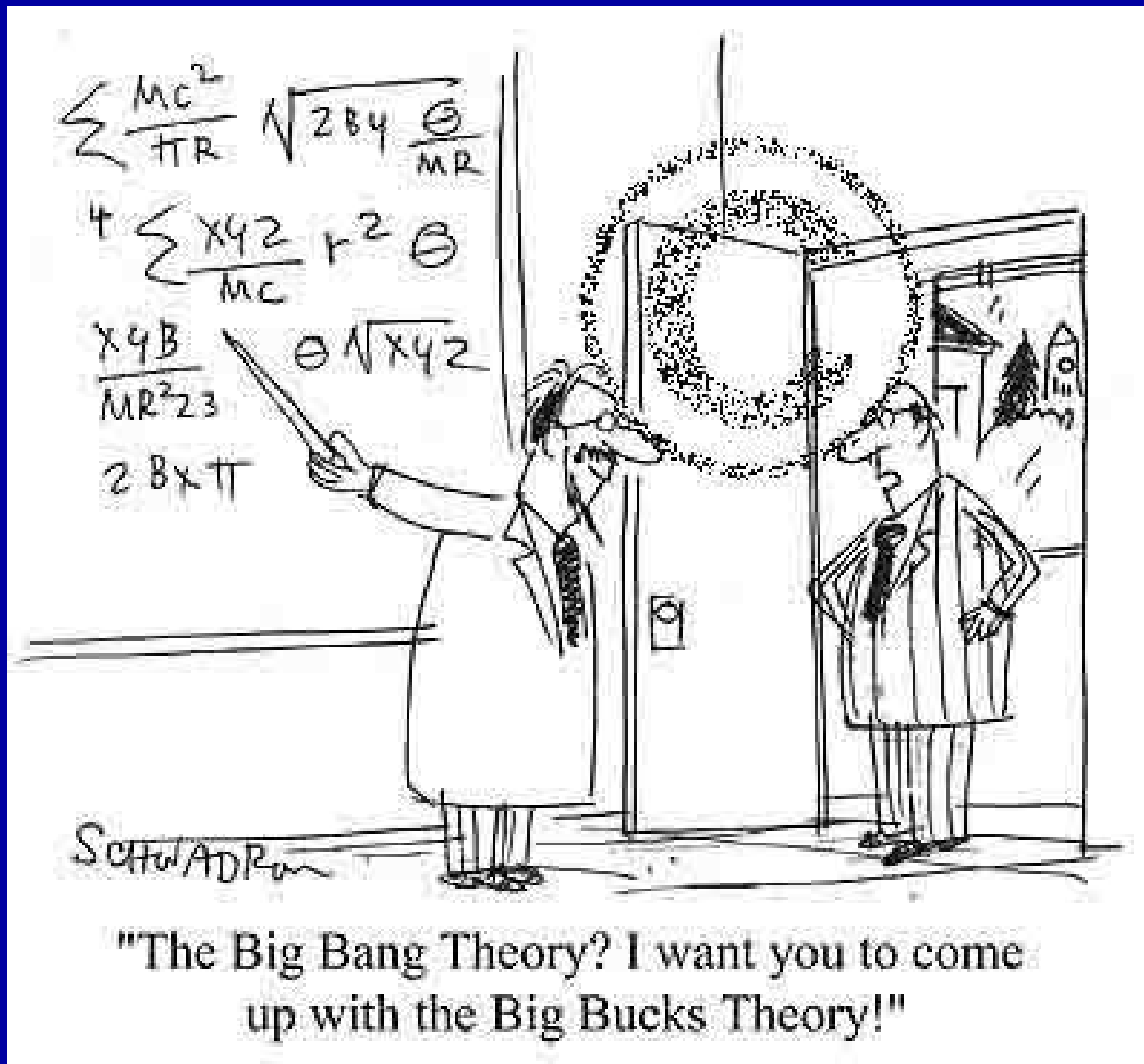
Using Hubble Space Telescope Galaxy Classification Software to Find Diabetic Type 2 in an Early Stage

Rogier Windhorst (Physics & Astronomy, ASU)

L. Burnett, J. Brower, A. Casano, D. Cox, J. Olson (ASU), R. & J. Herman (Banner Health), K. Tamura, V. Taylor (ASU)



Palm Grant proposal review — Banner Health — Phoenix, Aug. 30, 2007



Some cosmologists at ASU decided to step out of the box ...
(although it is hard to please the dean ...)

Outline

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) NASA missions to study Cosmic Dawn, First Light, Dark Ages
- (1b) NASA Hubble Space Telescope image analysis

(2) Using Galaxy Classification Software to Find Diabetic Type 2 in an Early Stage

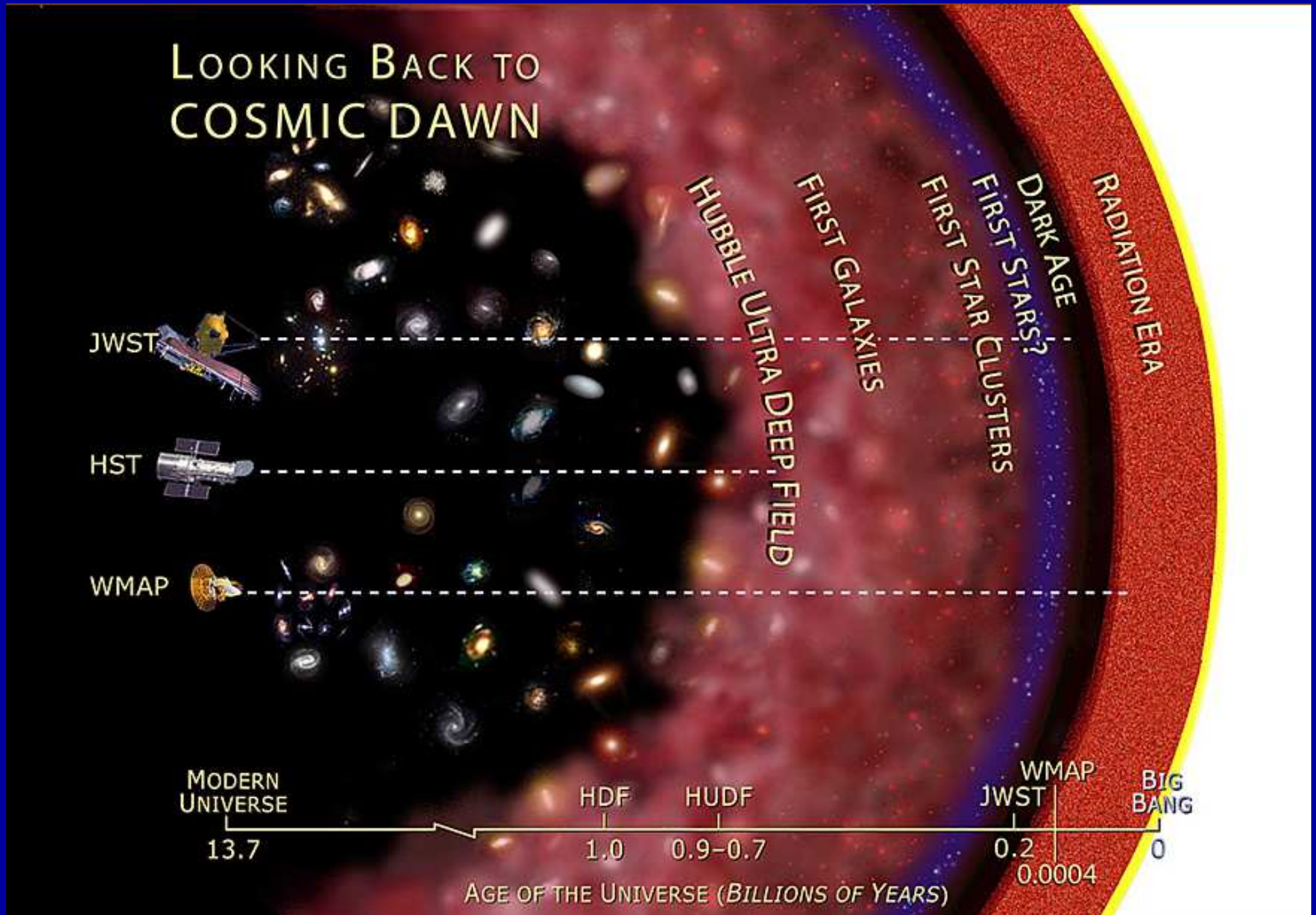
- (2a) The Galaxy Detection, Deblending, and Classification method
- (2b) Application to C-fiber Skin-biopsies to Normal and Obese Patients

(3) Results

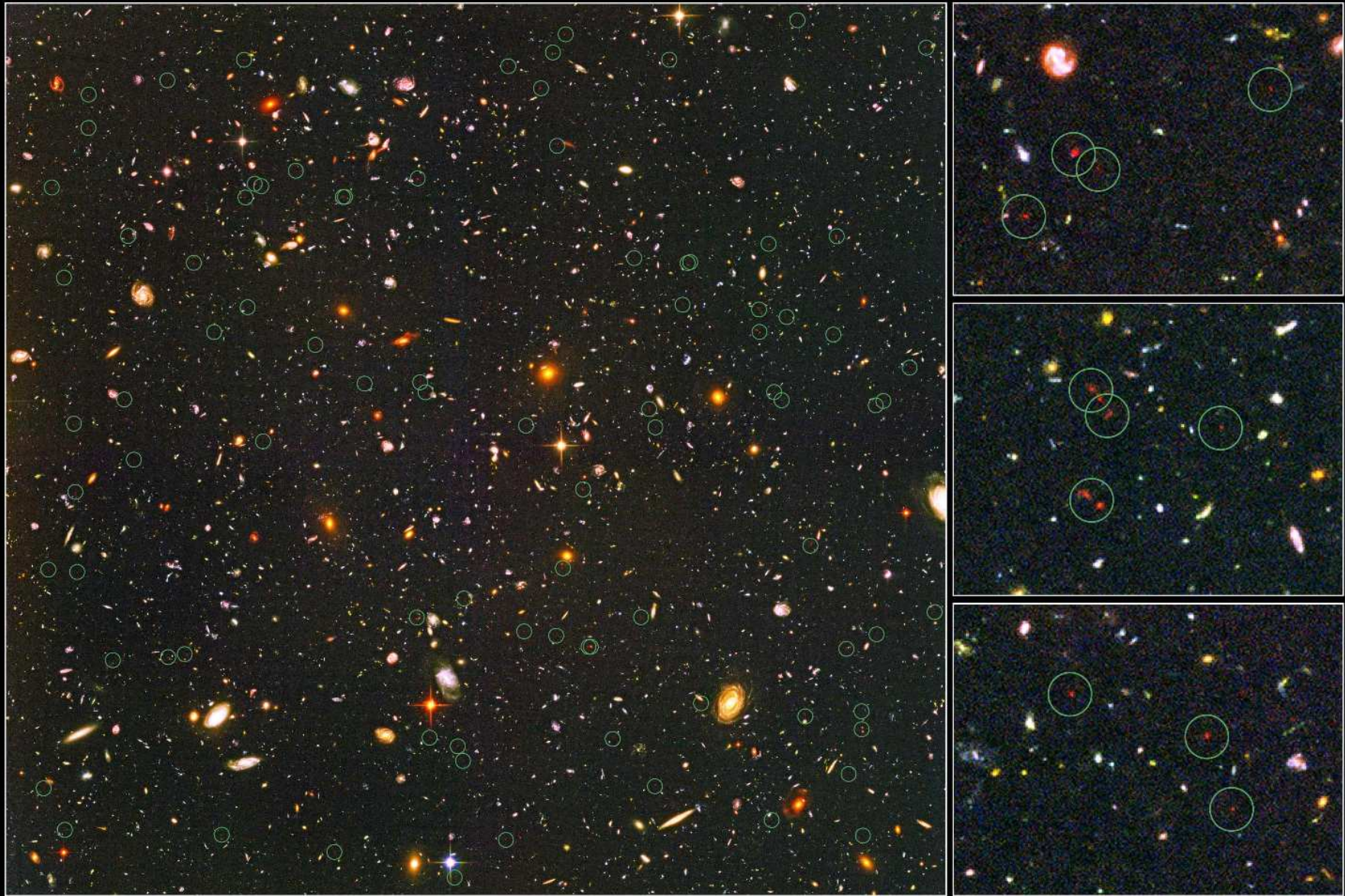
(4) Conclusions

Sponsored by NASA

(1a) NASA missions that study Cosmic Dawn, First Light, & the Dark Ages



Looking back in time $\implies$ deal with many overlapping galaxy images

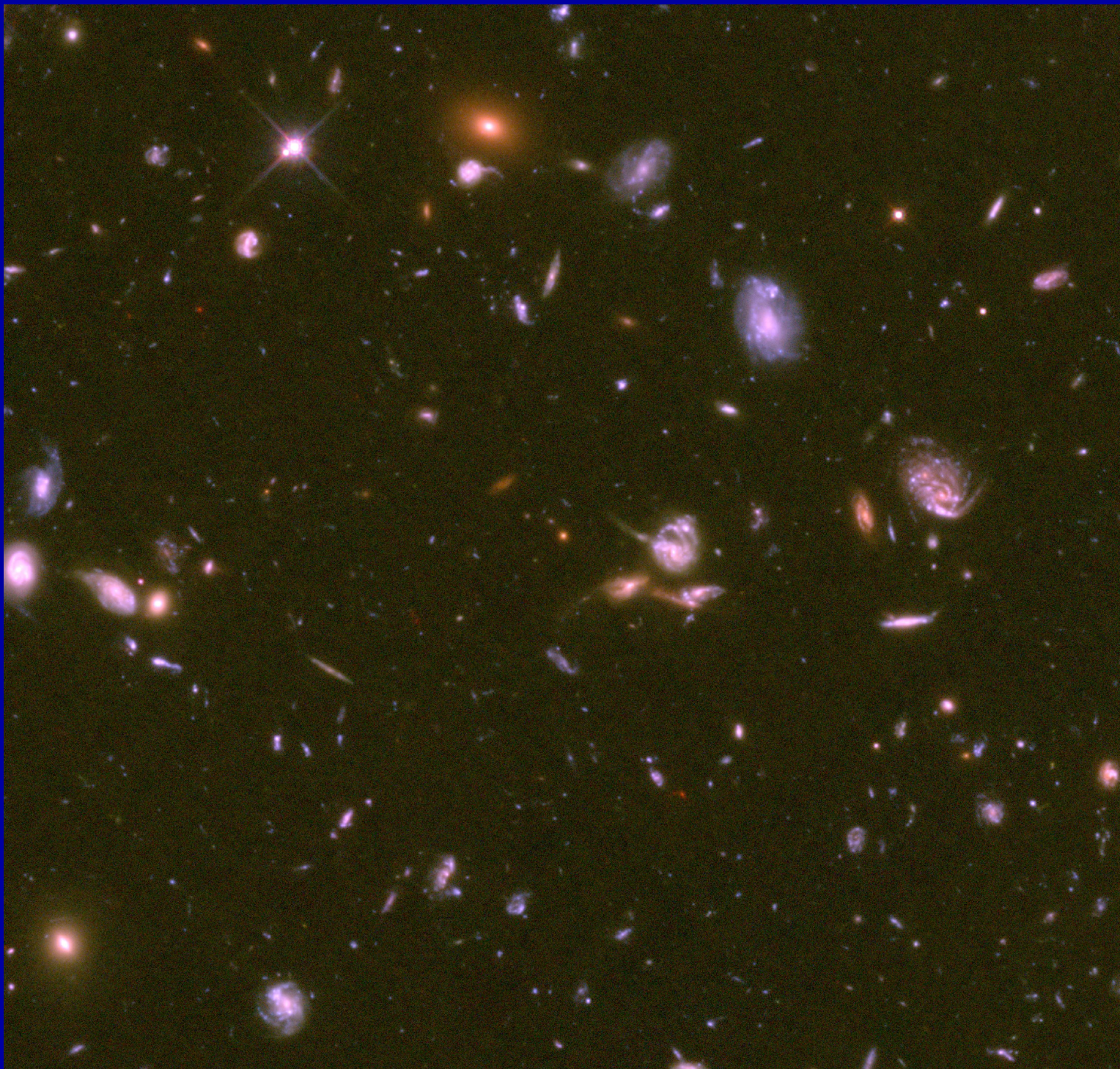


Distant Galaxies in the Hubble Ultra Deep Field
Hubble Space Telescope • Advanced Camera for Surveys

NASA, ESA, R. Windhorst (Arizona State University) and H. Yan (Spitzer Science Center, Caltech)

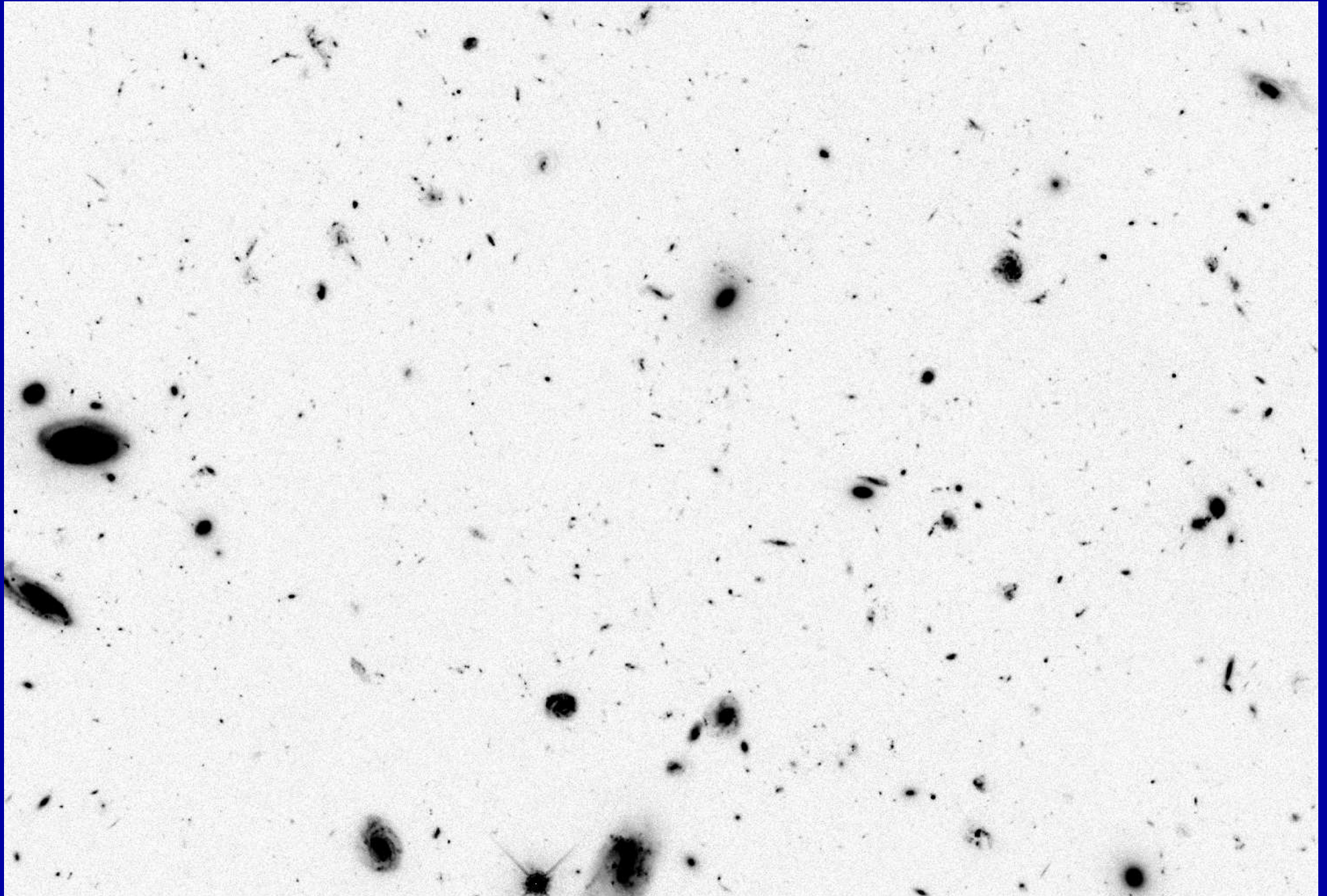
STScI-PRC04-28

A Recent Study of Cosmic Dawn with the Hubble Space Telescope



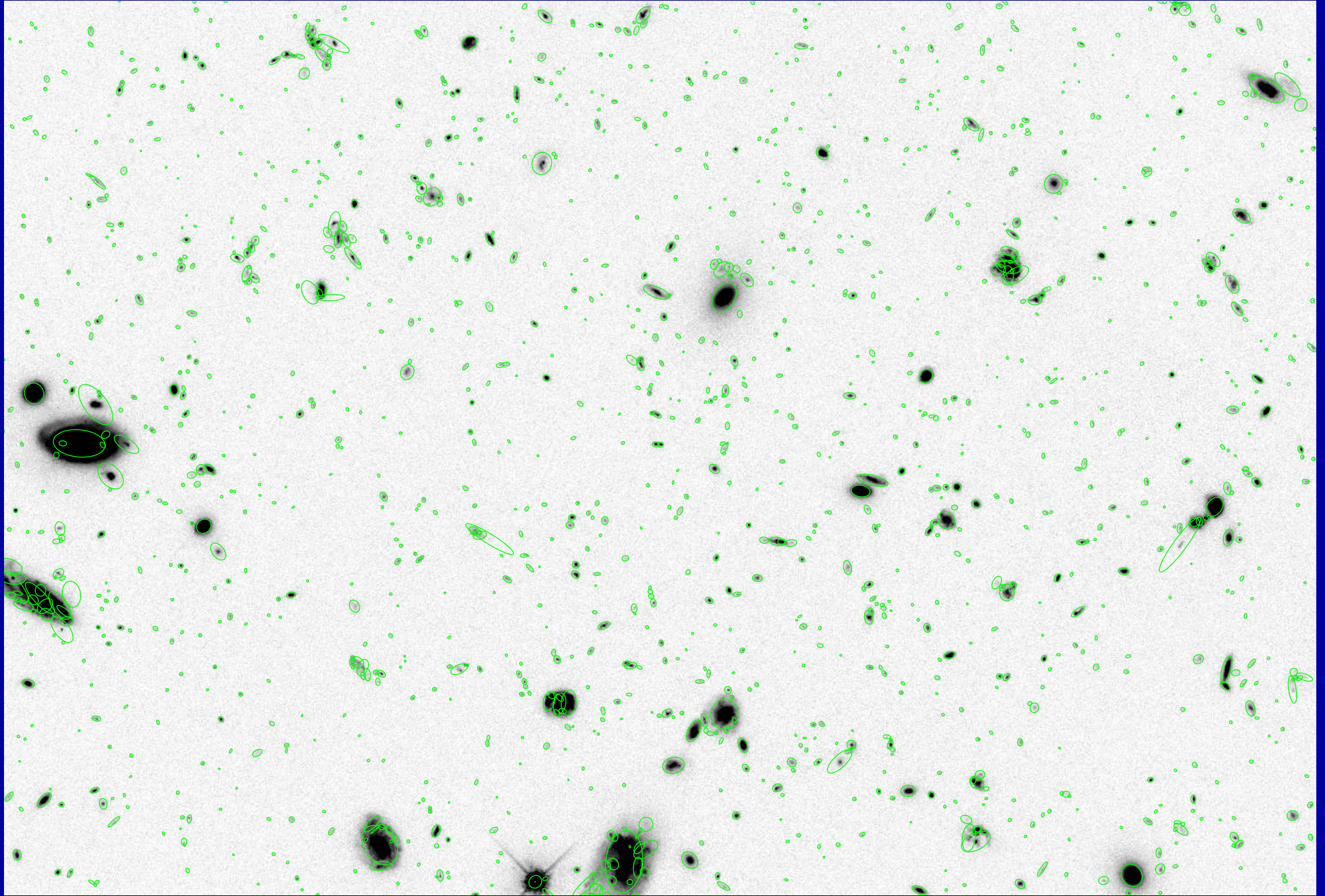
240 hours Hubble image in the Hubble UltraDeep Field (HUDF)

(1b) NASA Hubble Space Telescope image analysis



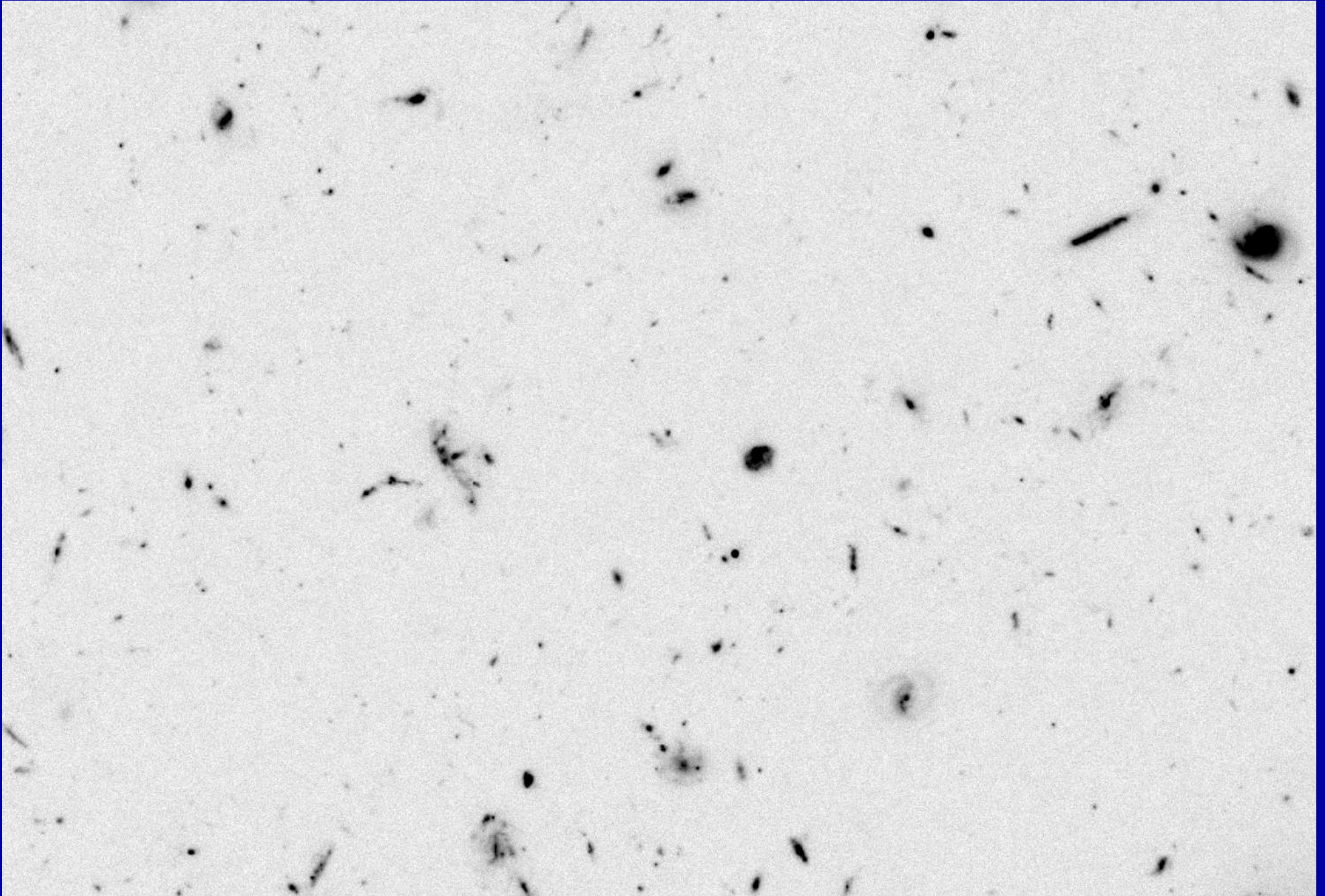
The deepest Hubble images start showing significant galaxy overlap.

(1b) NASA Hubble Space Telescope image analysis



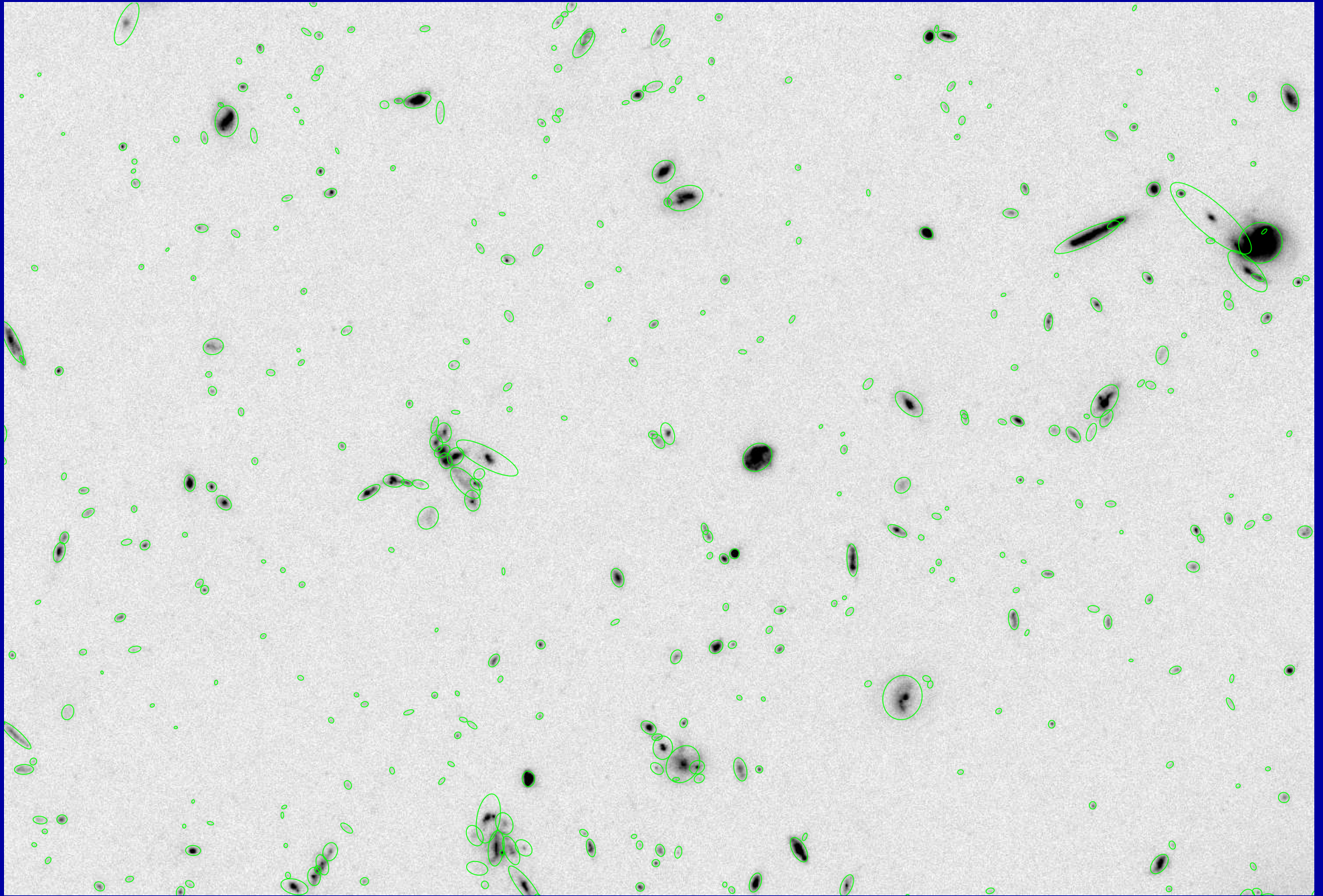
to address crowding, the source-extractor code iteratively deblends the images.

(2a) The Galaxy Detection, Deblending, and Classification method



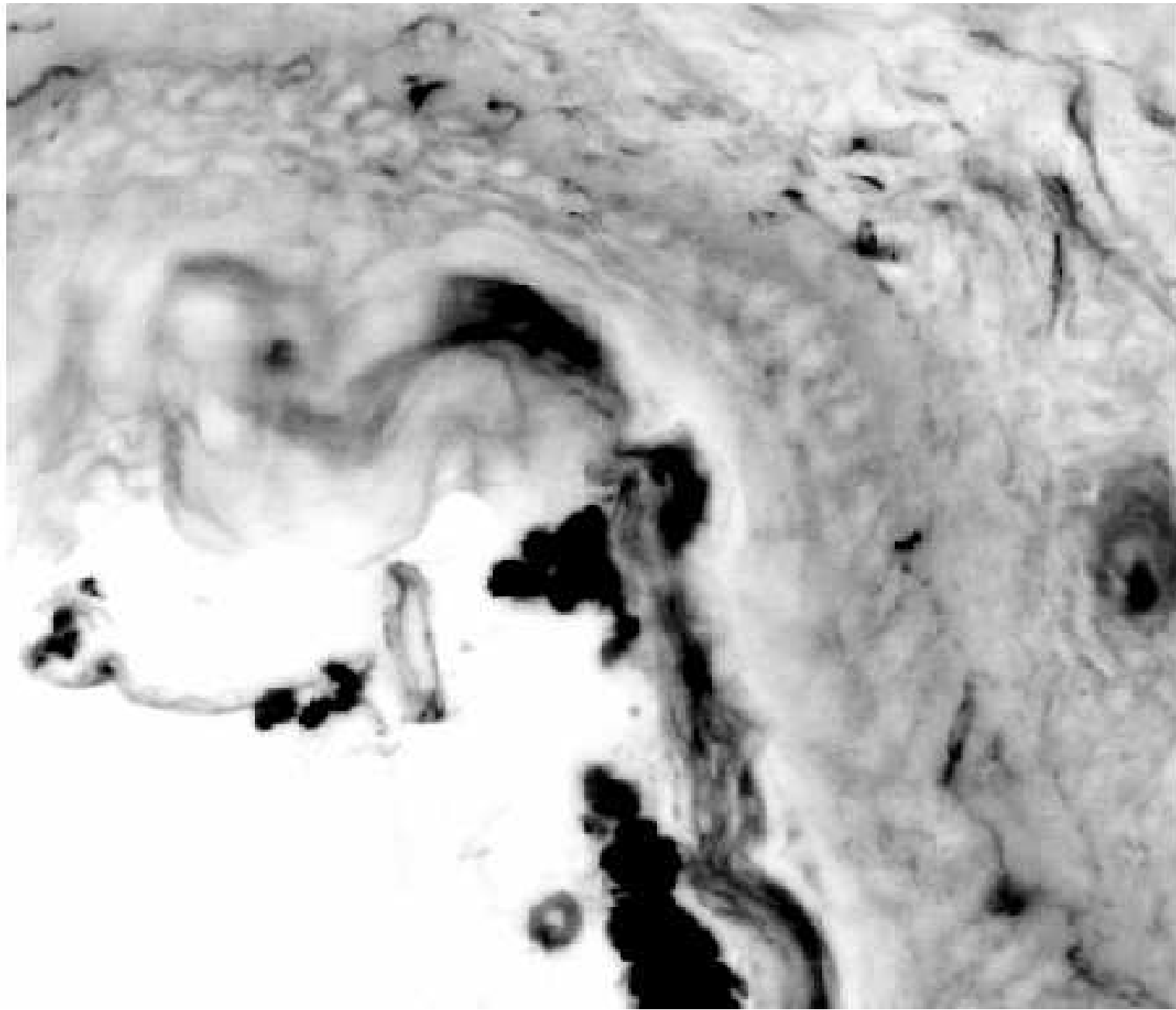
Very complex filamentary galaxies are seen in early stages of formation.

(2a) The Galaxy Detection, Deblending, and Classification method



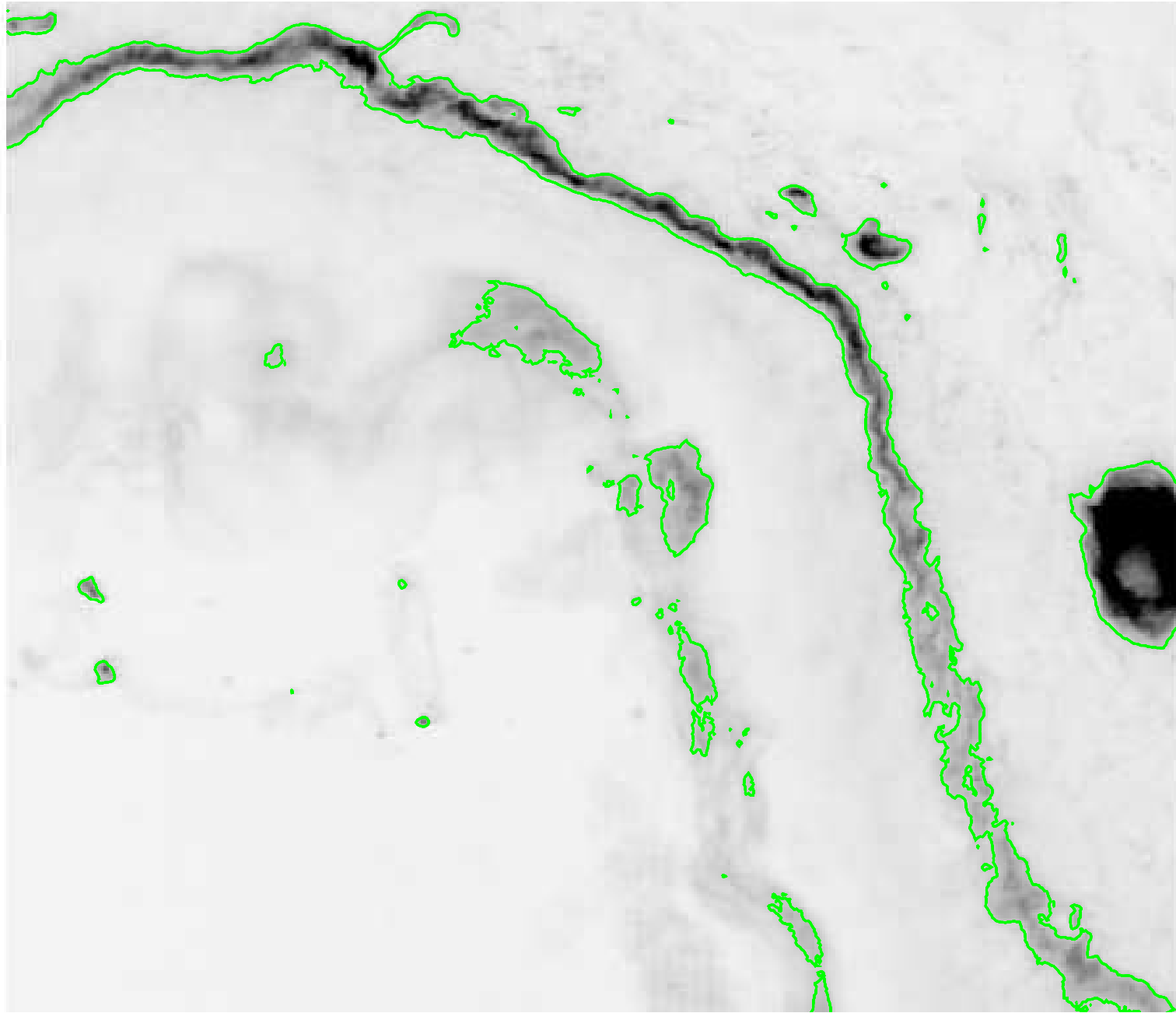
The Source-Extractor software deblends these in chains of elliptical isophotes.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



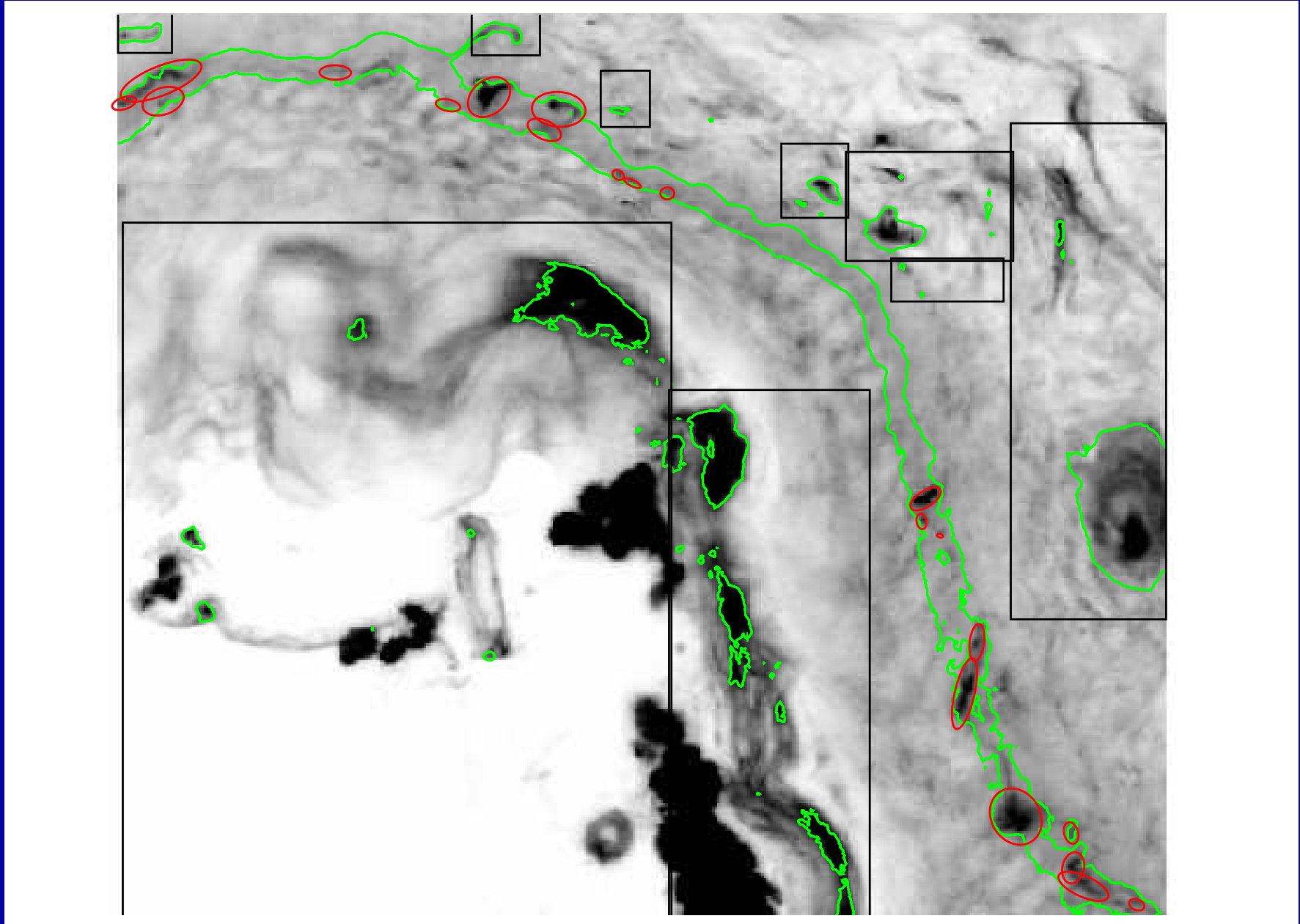
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



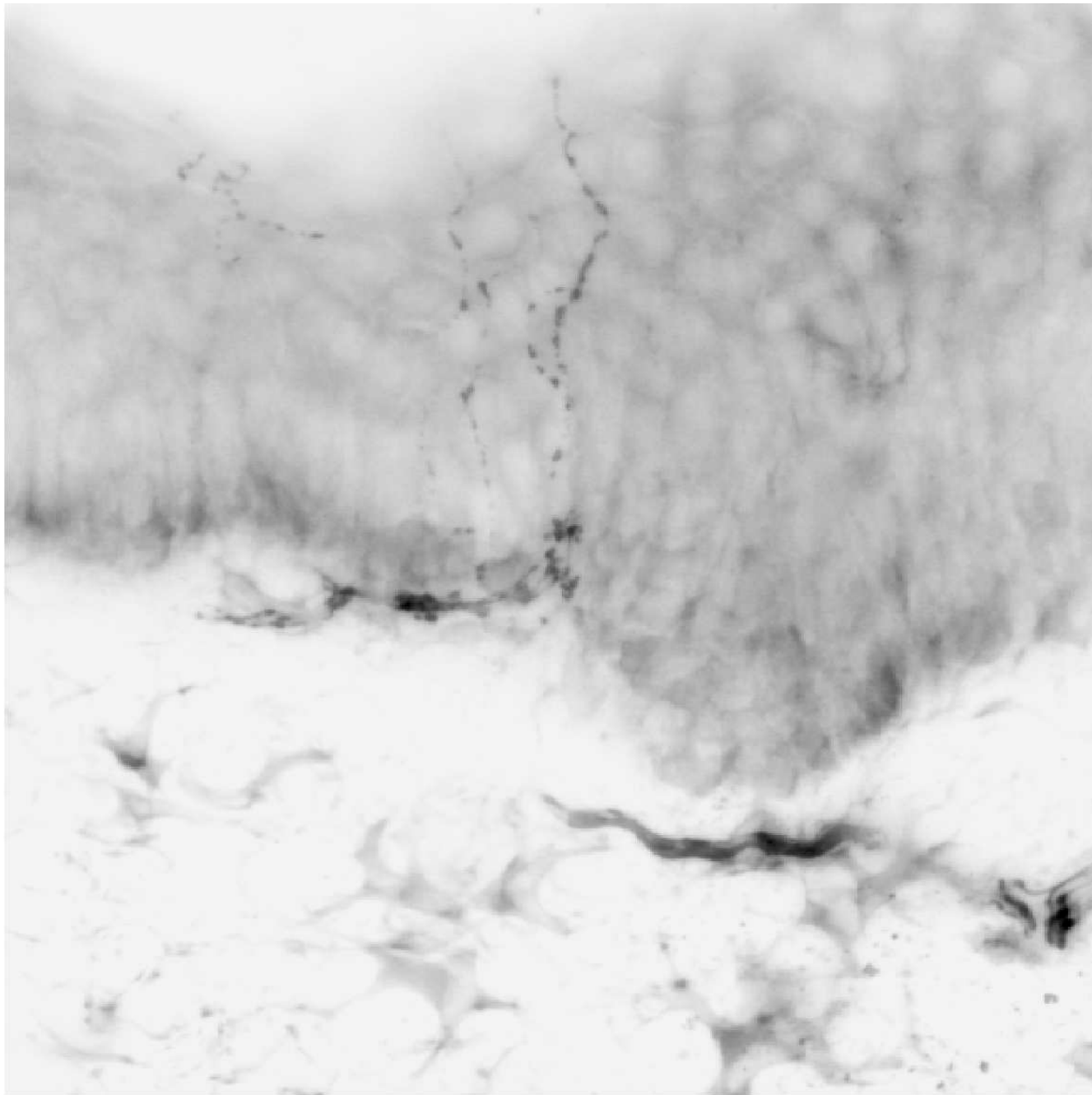
Different staining allows us to define the membrane

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



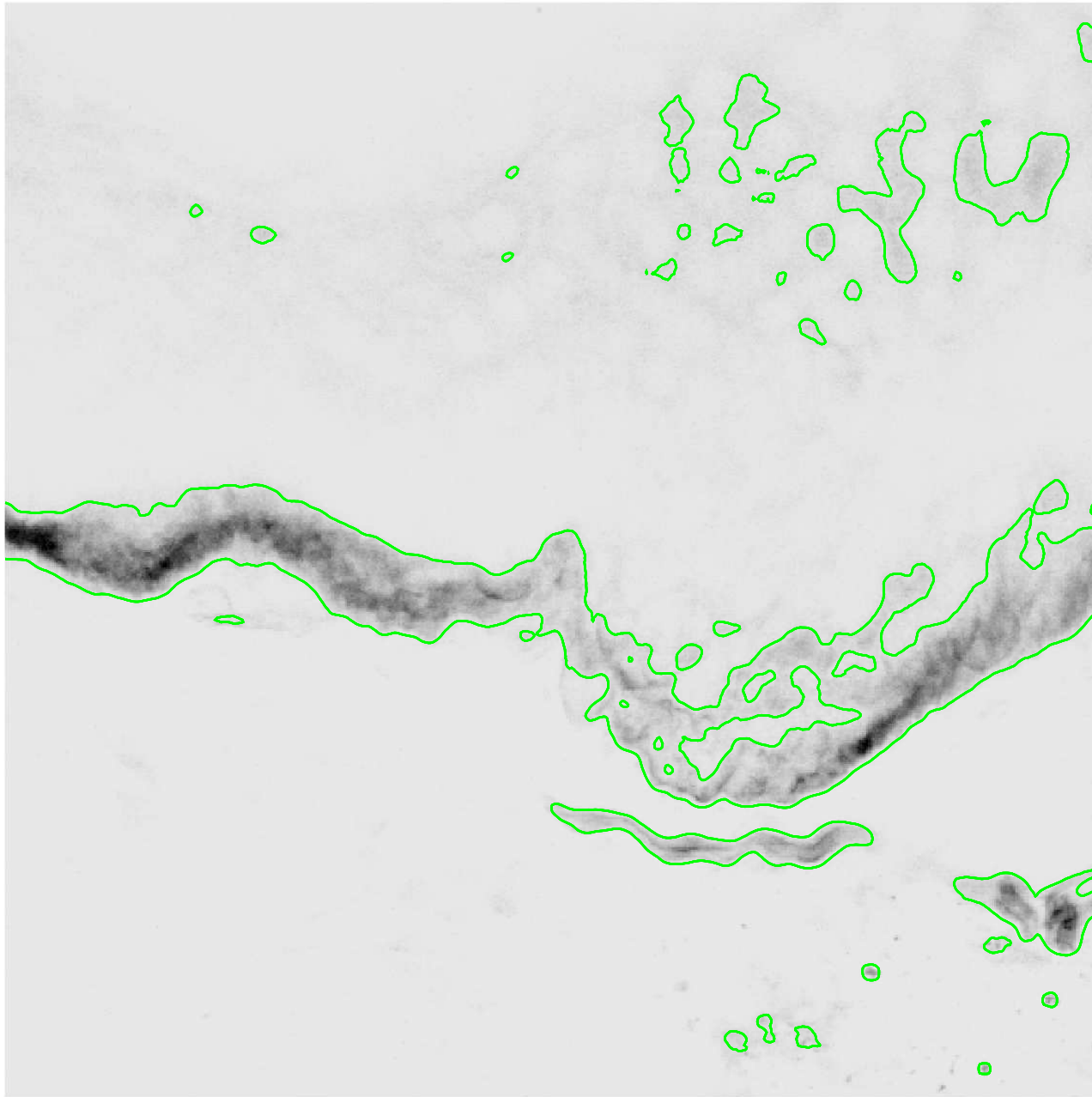
Source-Extractor finds C-fibers that cross the membrane (in red).
Black boxes: exclusion zone where Source-Extractor avoids bogus objects.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



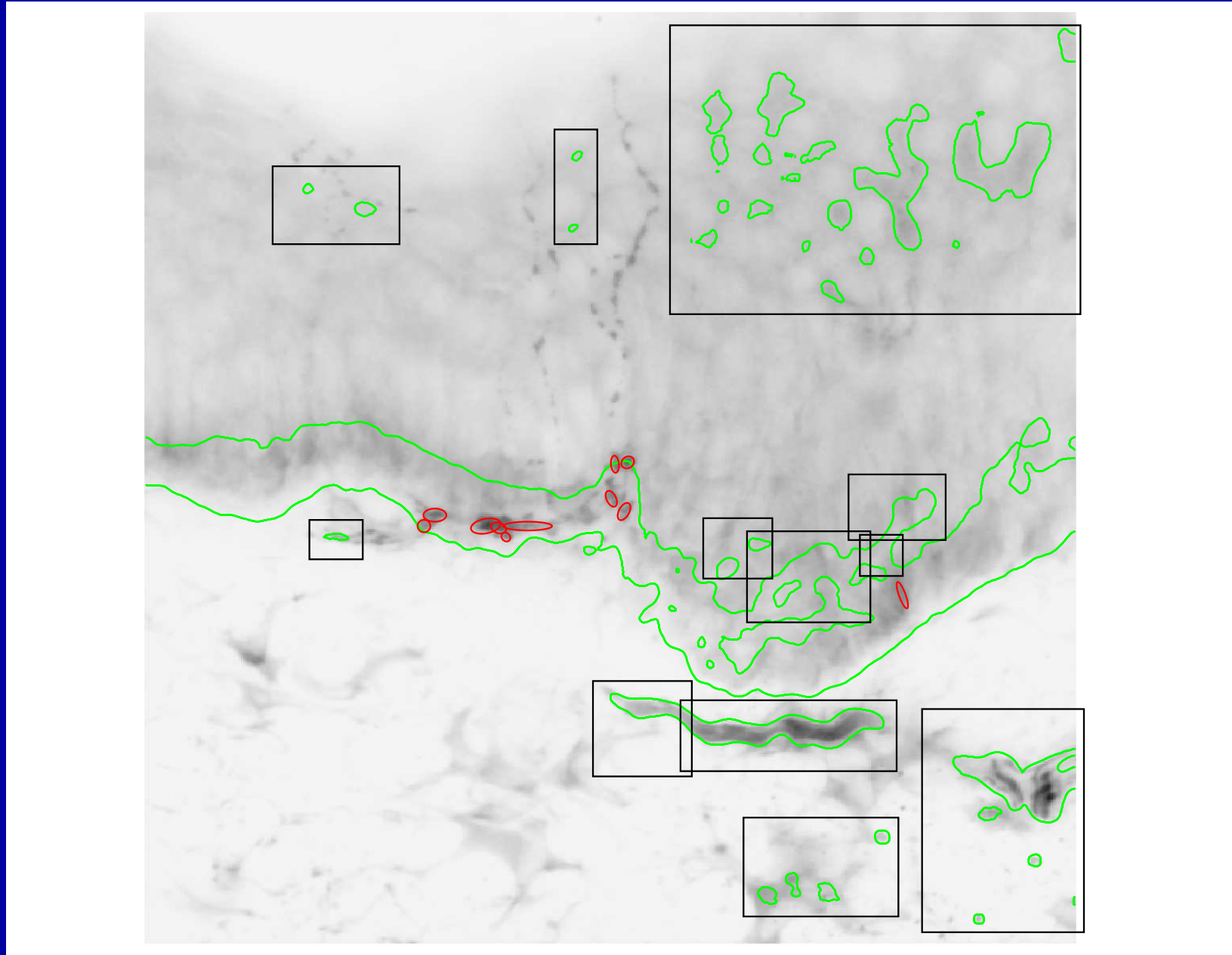
A Stained Confocal Microscope image of a Skin-Biopsy is digitized.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



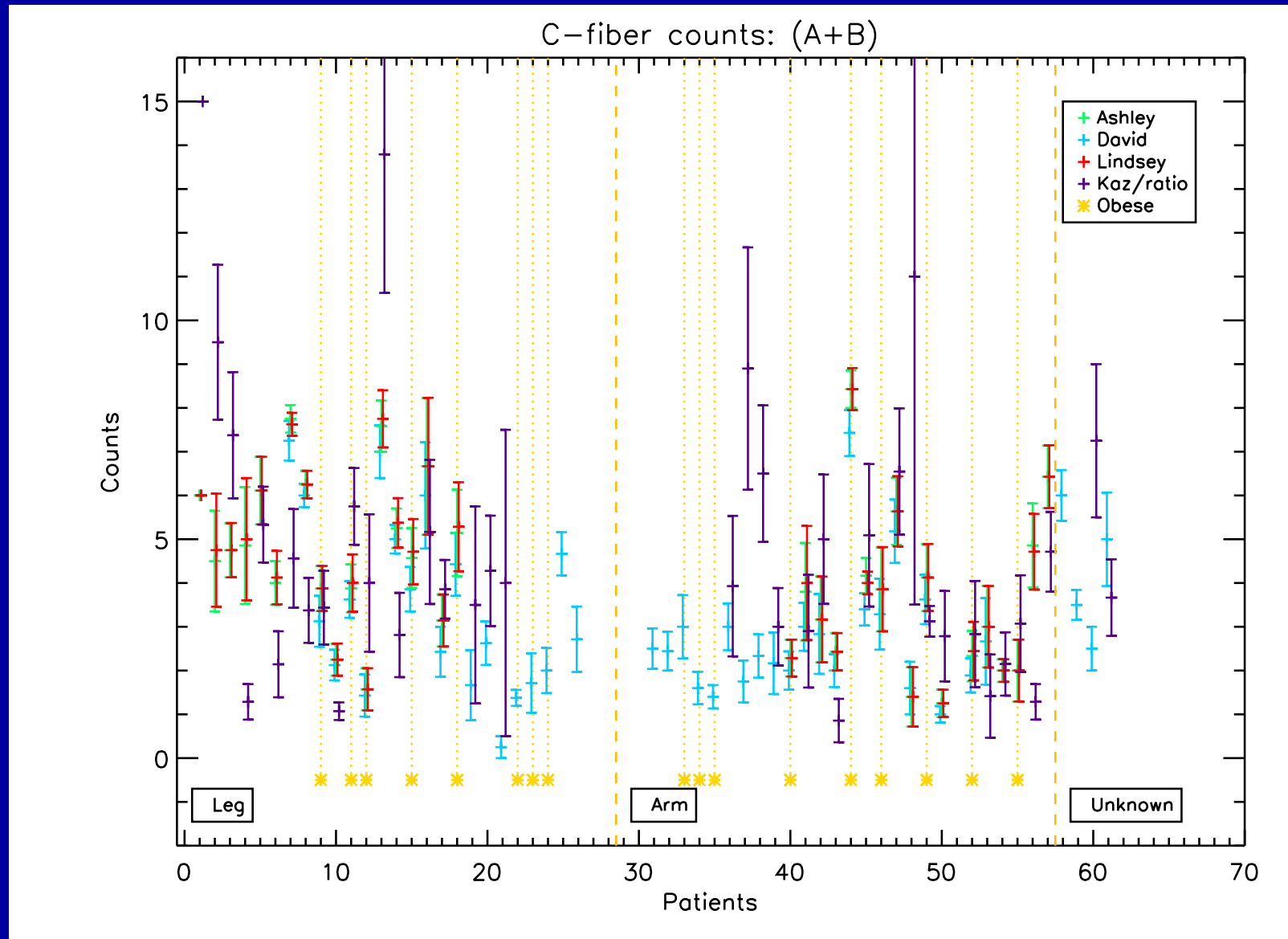
Different staining allows us to define the membrane.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



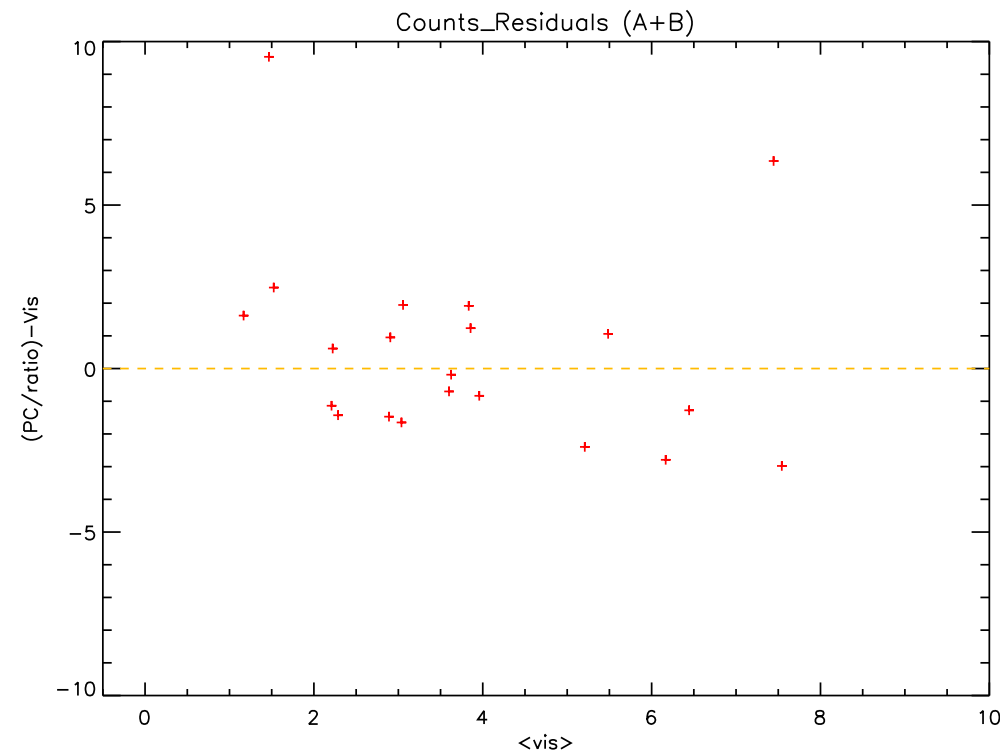
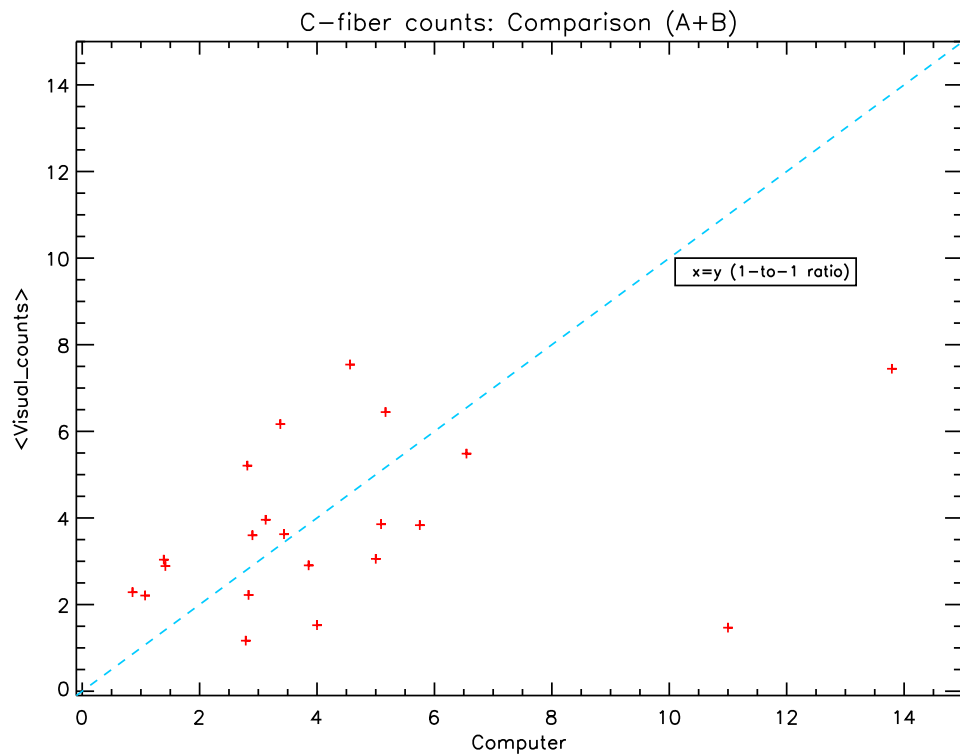
Source-Extractor finds C-fibers that cross the membrane (in red).
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(2b) Application to C-fiber Skin-biopsies



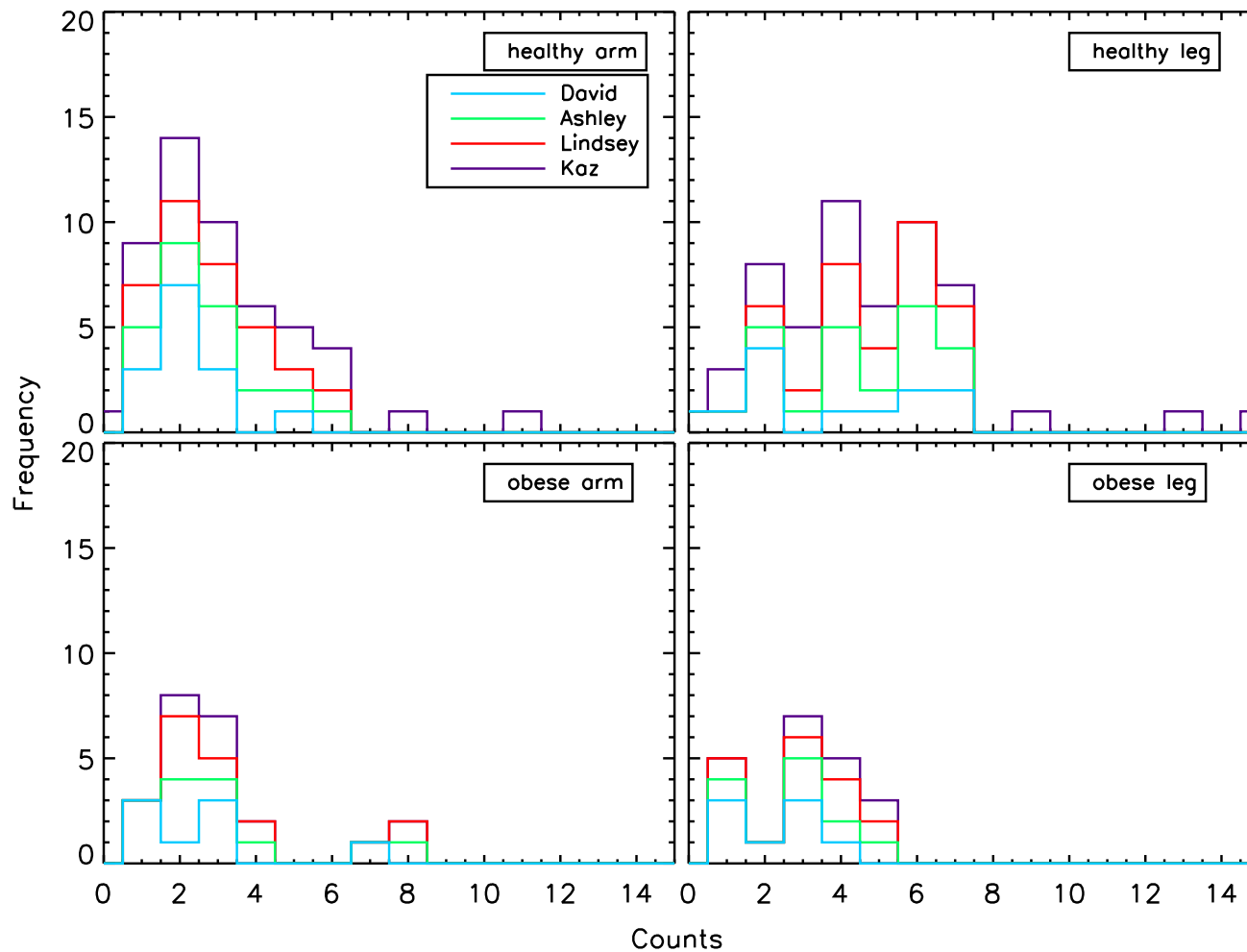
- Machine classification (Kaz — normalized) follows the 3 independent visual classifications quite well (vis counts not yet complete).
- Obese patients are in yellow.

(2b) Application to C-fiber Skin-biopsies



- Machine count correlates well with visual counts, with a few catastrophic failures where software breaks C-fiber up in too many pieces.
- Dispersion between and visual classifiers is $\sigma \simeq \pm 1.5$ C-fiber count.

(2b) Application to C-fiber Skin-biopsies



- Cumulative counts of both the machine and the visual classifiers show different distributions for normal and obese patients.
- Obese patients have in general lower C-fiber counts, although the differences between the two populations are not always huge.

(3) Conclusions

(1) Summary of the NASA Hubble Space Telescope connection

- (1a) Astronomers have no choice but to study 99.99999999999999999999999999990 % of the universe with clever software.
- (1b) NASA Hubble Space Telescope image analysis software can detect, deblend and classify faint galaxy or faint star images.
- (1c) Software is very general — doesn't care what it counts or classifies!

(2) Can use classification software to find Diabetic Type 2 in early stage:

(2a) Galaxy Detection, Deblending, and Classification software is fairly robust and general, minimizing human interaction.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese patients allows us to quantitatively distinguish healthy from sick patients: Dispersion with visual classification is $\sigma \simeq \pm 1.5$ C-fiber count.

(2c) More statistics are definitely needed to better cover subsamples of: healthy, obese, adult, child, and rare classes.

SPARE CHARTS

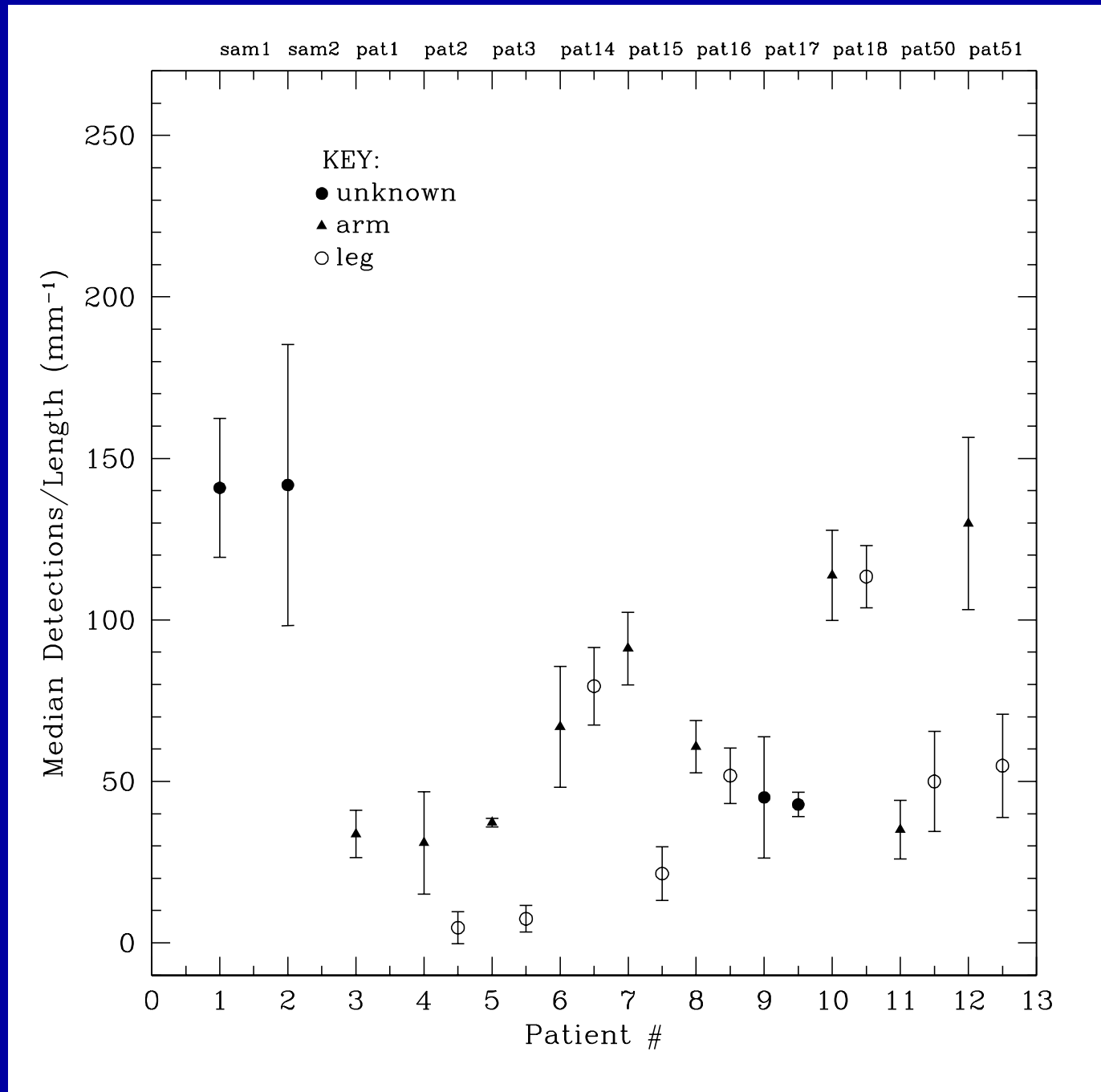
(1) The NASA Hubble Space Telescope connection

- [illegible]

⇒ Astronomers must be smart about using software to analyze what they see in Hubble images — we have absolutely no alternatives!.

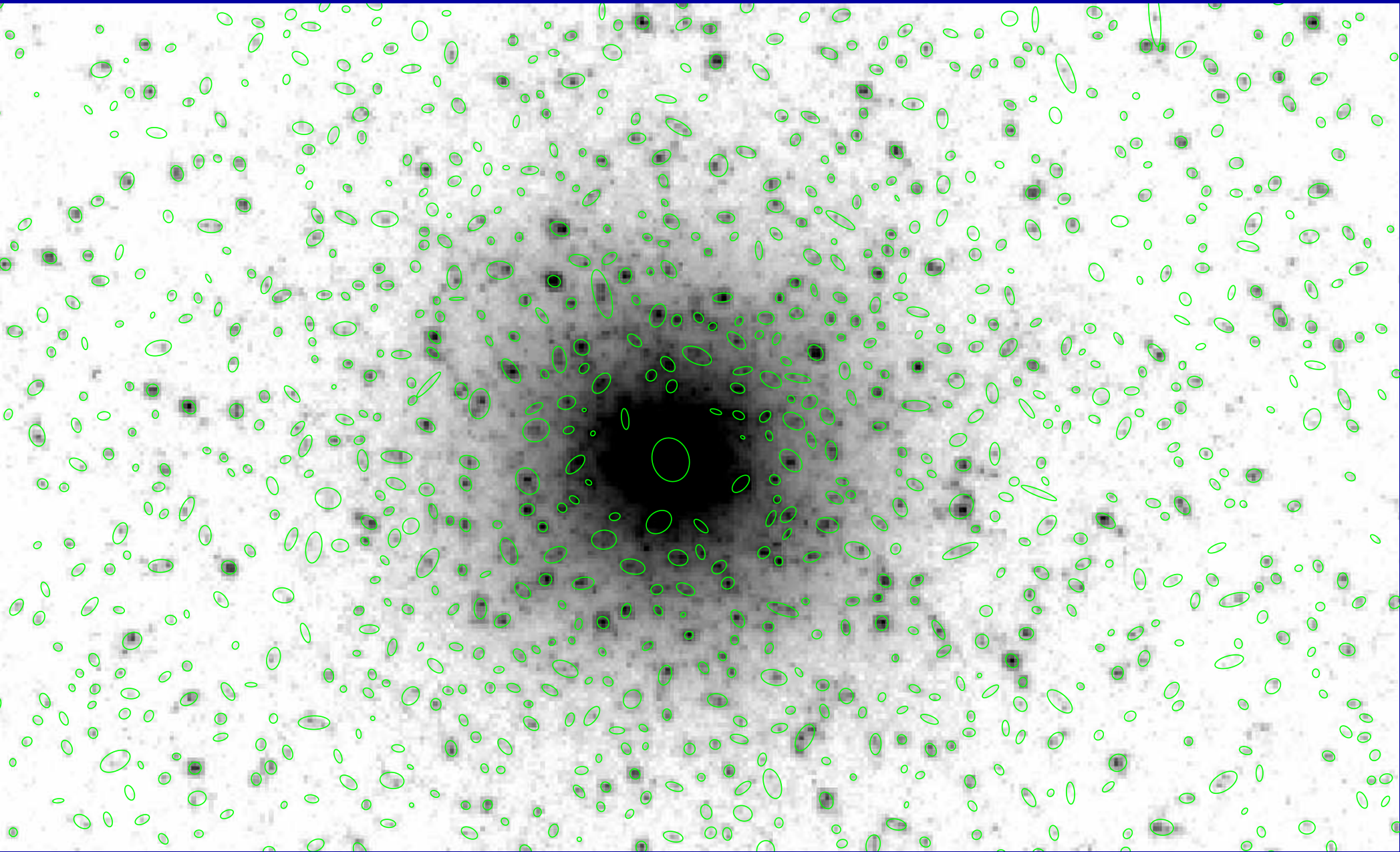
This project of medical applications of Hubble galaxy classification software builds on a 40 year history of astronomical software development.

(2b) Application to C-fiber Skin-biopsies in Normal and Obese Patients



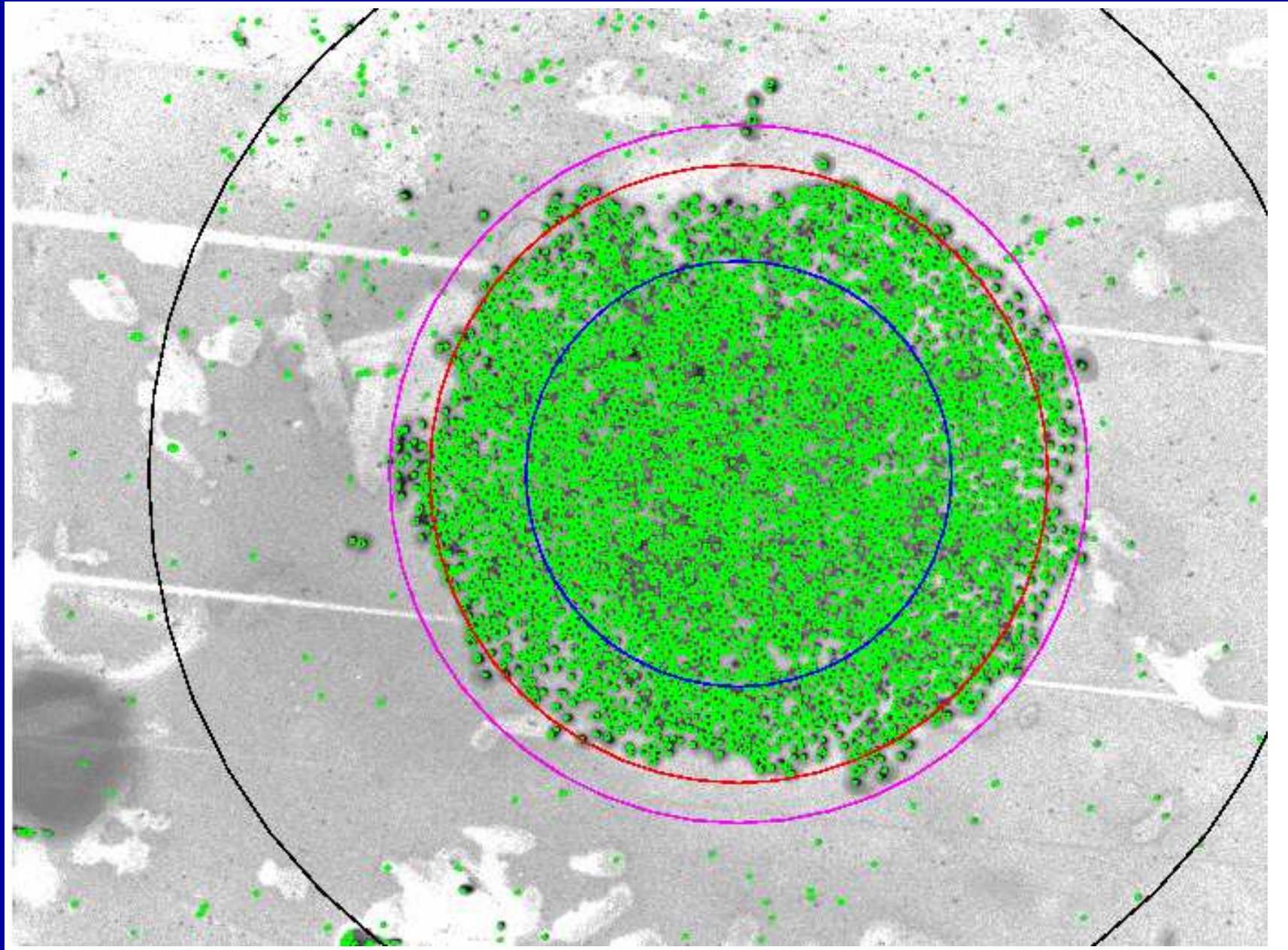
Number of C-fibers crossing membrane is directly counted for each patient.

(2a) The Galaxy Detection, Deblending, and Classification method



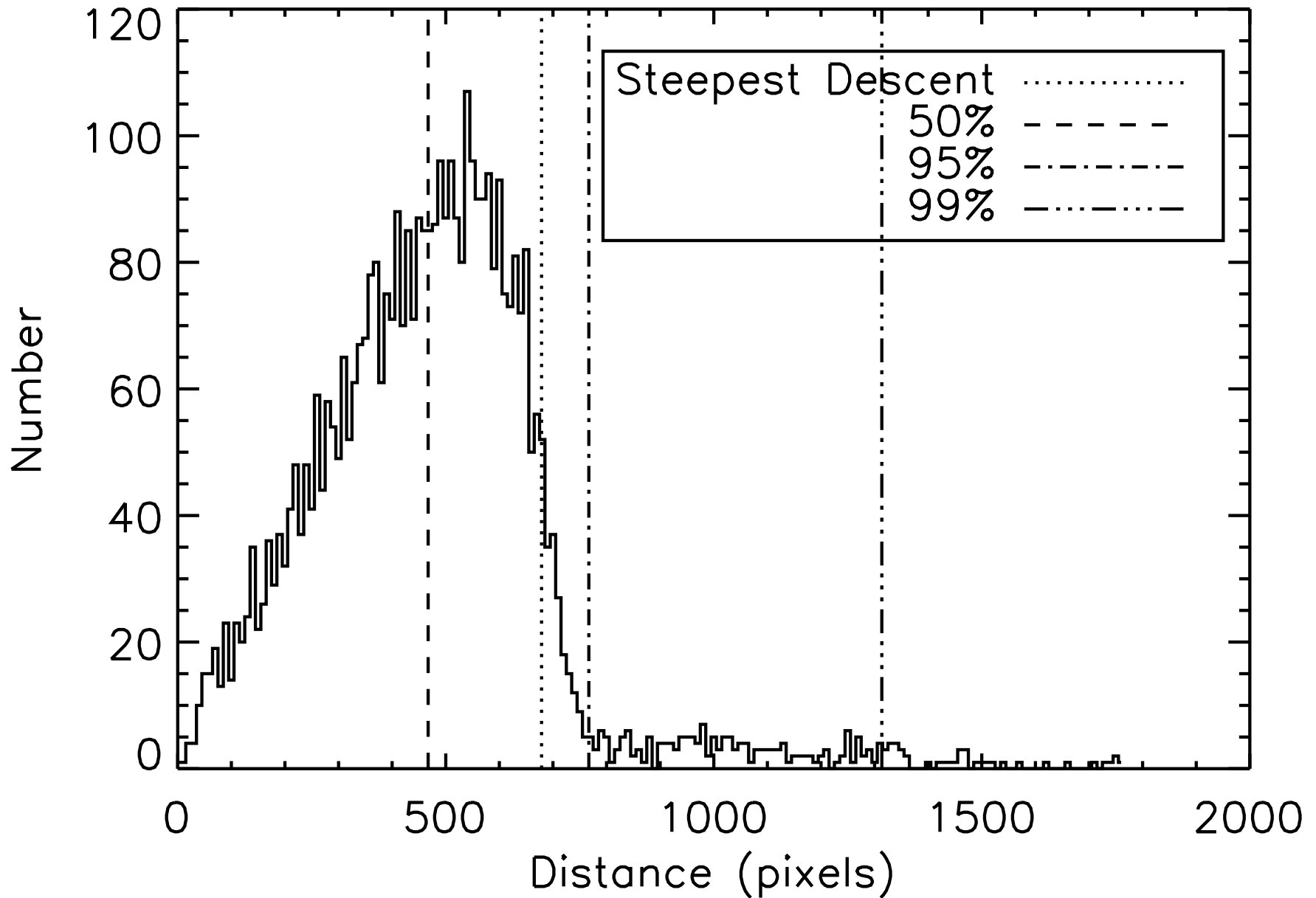
The Source-Extractor algorithm deblends a rich globular cluster of 10^5 stars.
(Only several 10^3 brightest stars plotted, but object overlap is allowed).

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



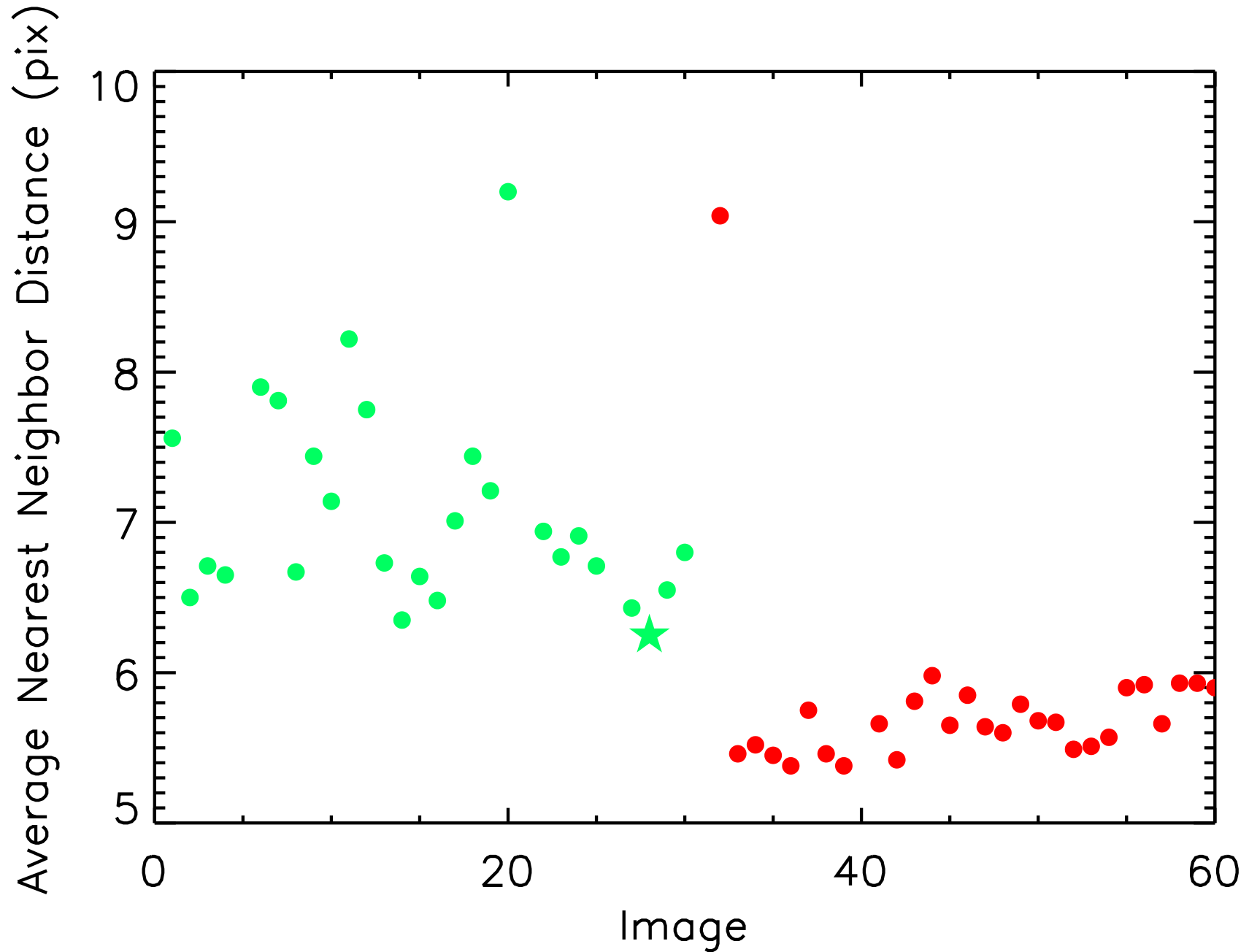
Source-Extractor directly counts number of cells in a cancer biopsy image.
(Collaboration with Dr. M. Berens et al. at TGEN, R. Ryan at ASU.)

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



The distribution and spreading of cancer cells is measured quantitatively.

(2c) Application of Hubble Software to Study Spreading of Cancer Cells



Nearest neighbor analysis bifurcates the spread and non-spread cancer biopsies.